

Handheld LiDAR mapping

This feature requires launching the program inside the `slam_gmapping` package, which is located in the **yahboomcar_ws** source package. The following instructions use the bundled virtual machine as an example. If you intend to run this on your own board, place `yahboomcar_ws` in the home directory and build it there.

1. Launch the LiDAR

Open a terminal and run:

```
ros2 launch ydlidar_ros2_driver ydlidar_launch.py
```

2. Publish Static Odom Transform

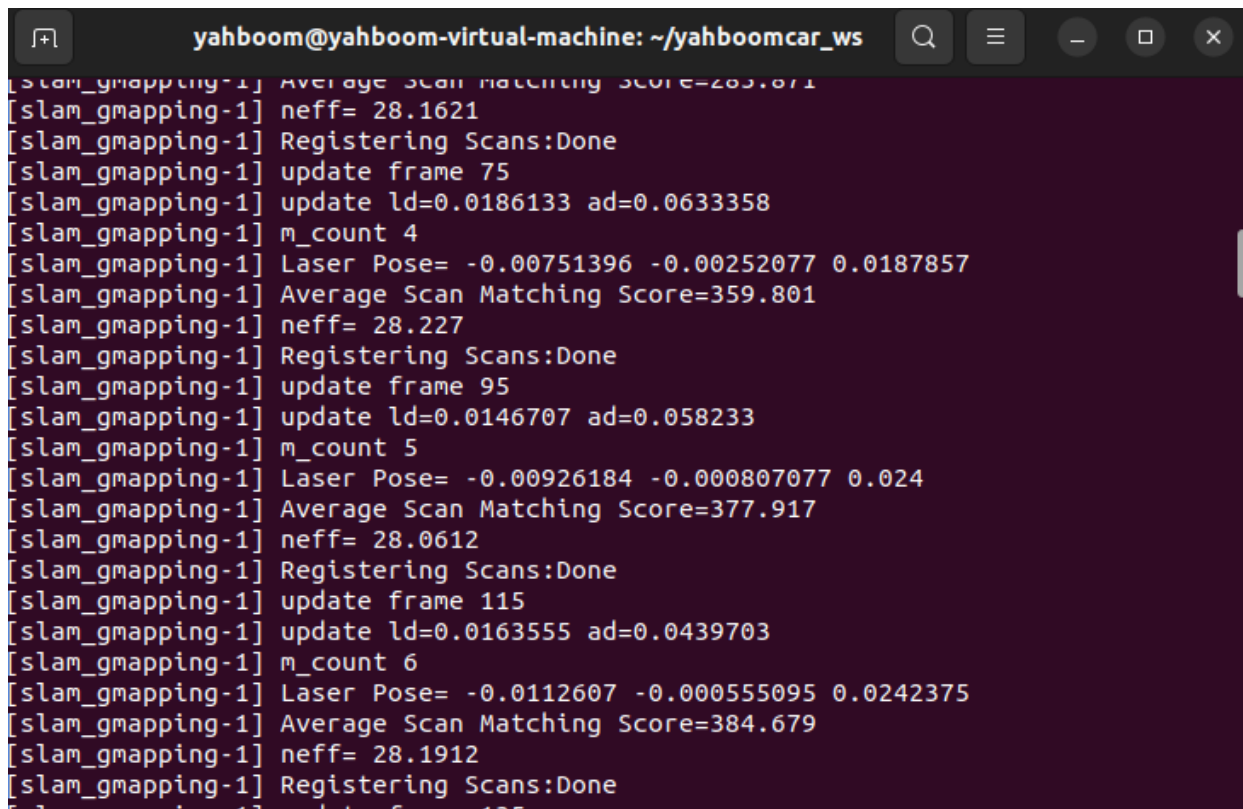
Open a terminal and run:

```
ros2 launch rf2o_laser_odometry rf2o_laser_odometry.launch.py
```

3. Start GMMapping

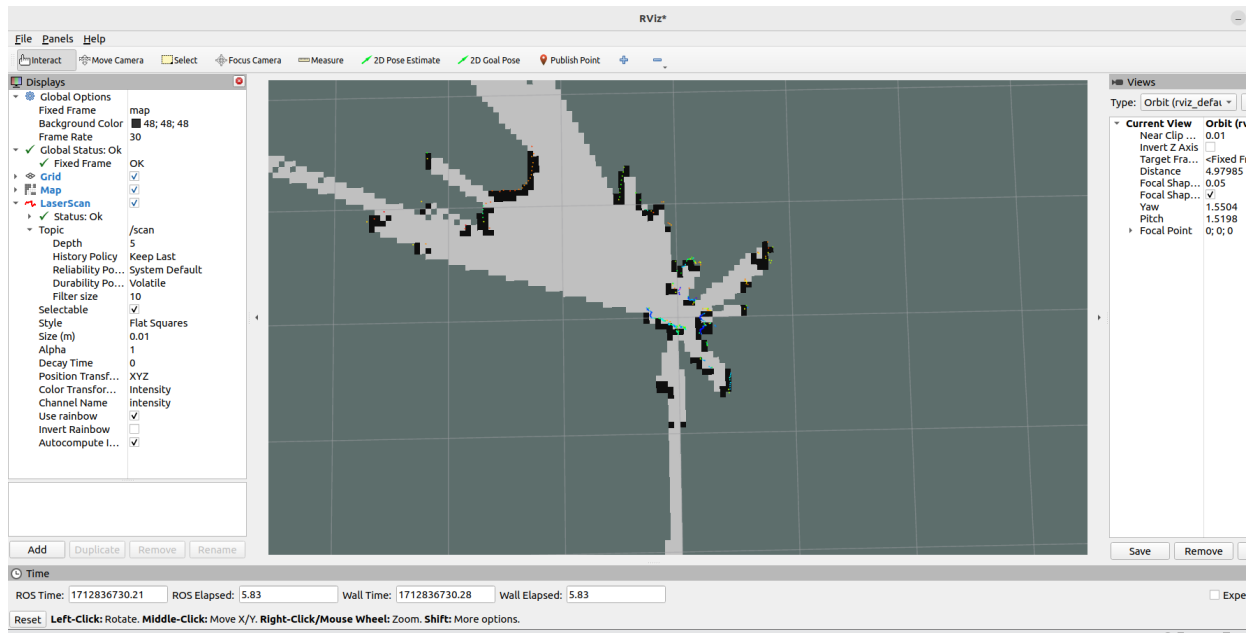
Open a terminal and run:

```
ros2 launch slam_gmapping slam_gmapping.launch.py
```



The RViz display should appear as shown in the screenshot.

If no map is displayed, open the RViz map configuration manually by clicking: **File** → **Open Config** → **sunrise** → **yahboomcar_ws** → **src** → **slam_gmapping** → **rviz** → **map_view.rviz**



4. View the TF Tree

Open a terminal and run:

```
ros2 run tf2_tools view_frames
```

```
yanboom@yanboom-virtual-machine:~/yahboomcar_ws$ ros2 run tf2_tools view_frames
[INFO] [1712836863.067737751] [view_frames]: Listening to tf data for 5.0 second
s...
[INFO] [1712836868.112822280] [view_frames]: Generating graph in frames.pdf file
...
[INFO] [1712836868.118691540] [view_frames]: Result:tf2_msgs.srv.FrameGraph_Response(frame_yaml="odom: \n parent: 'map'\n broadcaster: 'default_authority'\n
rate: 20.197\n most_recent_transform: 1712836868.110051\n oldest_transform: 1712836863.059761\n buffer_length: 5.050\nlaser_frame: \n parent: 'base_link'\n
broadcaster: 'default_authority'\n rate: 10000.000\n most_recent_transform: 0.000000\n oldest_transform: 0.000000\n buffer_length: 0.000\nbase_link: \n pa
rent: 'base_footprint'\n broadcaster: 'default_authority'\n rate: 10000.000\n
most_recent_transform: 0.000000\n oldest_transform: 0.000000\n buffer_length: 0.000\nbase_footprint: \n parent: 'odom'\n broadcaster: 'default_authority'\n
rate: 10.237\n most_recent_transform: 1712836867.890208\n oldest_transform: 1712836863.005946\n buffer_length: 4.884\n")
vahboom@vahboom-virtual-machine:~/vahboomcar_ws$
```

This will generate a **frames.pdf** file in the directory where the command was executed. That file contains the generated TF tree.

view_frames Result

Recorded at time: 1712836868.1316395

